

Answers

1. RGB coordinates in RGB space are equivalent to Cartesian coordinates in Cartesian space.

Answer: True

2. RGB space interpolation always matches our perceptual expectations.

Answer: False

3. Interpolating between two colours in a non-RGB space and then converting back to RGB can produce better results.

Answer: True

4. Interpolation between (2, 0) and (4, 0) in polar coordinates results in linear movement in Cartesian space.

Answer: True

5. Polar coordinates are an example of a non-Cartesian space.

Answer: True

6. Using polar coordinates for interpolation can result in movement along the perimeter of a circle.

Answer: True

7. What might be some unintended consequences of using lerp?

Answer: While lerp can interpolate many values, the result might not always match perceptual or artistic expectations. For instance, interpolating between certain colours in RGB space might produce unwanted intermediate hues.

8. How do RGB values relate to Cartesian coordinates?

Answer: RGB values can be seen as 3D points in RGB space, akin to x, y, z axes in Cartesian coordinates.

9. What problem arises when trying to interpolate colours using the RGB space?

Answer: RGB space interpolation may not always match human perceptual expectations. As a result, when transitioning between colours you may end up with unexpected or undesirable intermediate colours.